# EXPERIMENT 7

## SLIDING WINDOW PROTOCOL.

AIM:

To write a program to implement flow control at data link layer using sliding window protocol stimulate the flow

Sender code:

```
import time
import os
Print ("=" * 55)
Print ("sliding window Protocol- Sender")
if (window_size <= 0;
    print ("Invalid window size!")
    exit()
message = input ("enter text message:")
if not message:
    print ("Message cannot be empty")
    exit ()
try:
  with open ("Receiver_Buffer." "r") as file:
      response. file. readline (). strip()
  except fileNOTFound error:
      print ("Receiver_ buffer not found)")
  parts = response . split ()
  if len (parts): 2
      print ("Invalid Receiver. Buffer found!")
      continue.
```

```
    else:
        print("Invalid ack response received time.sleep(1)
Print("All frames transmitted successfully")
Print("original message {message}")
```

Output:

```
Sliding window protocol Sender
Enter window size: 3
Enter test message: HELLO
```

Receiver Code:

```
import os
SENDER_FILE = "sender_buffer.txt"
RECEIVER_FILE = "receiver_buffer.txt"
expected_frame = 0
while True:
    if not os.path.exist(sender_file))
    print("\n frames received")
    error = false
    error.frame = None.

else:
    with.open(RECEIVER.file,'w') as file:
        file.write(f"ACK {expected file})
    print("\n All frames received correctly)
    print(f"ACK: {expected_frame}")
```

Output:

frames Received

frame No: 0 Data: H

frame No: 1 Data: E

frame No: 2 Data: L

All frames received Correctly

ACK: 3

RESULT:

Thus the sliding window protocol was implement successfully using Python and flow of frames btw sender and receiver was simulated using ACK/NACK.